



Application Descriptions

7

Energy Management

80

Photovoltaics

1

Summary

This document specifies the Functional Blocks for the control of photovoltaic appliances and energy management.

This document is WGI Approved.

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1 Application Model

The Application “Photovoltaics” consists of the following Functional Blocks.

- Inverter PV control for generation of solar PV electric DC power.
- Inverter Battery Control for storage of solar PV electric DC power.
- DC-AC inverter monitor for DC-to-AC conversion between DC power and the AC power distribution network.
- PCC Monitor for information about “power consumption” and “feed in” to the grid.

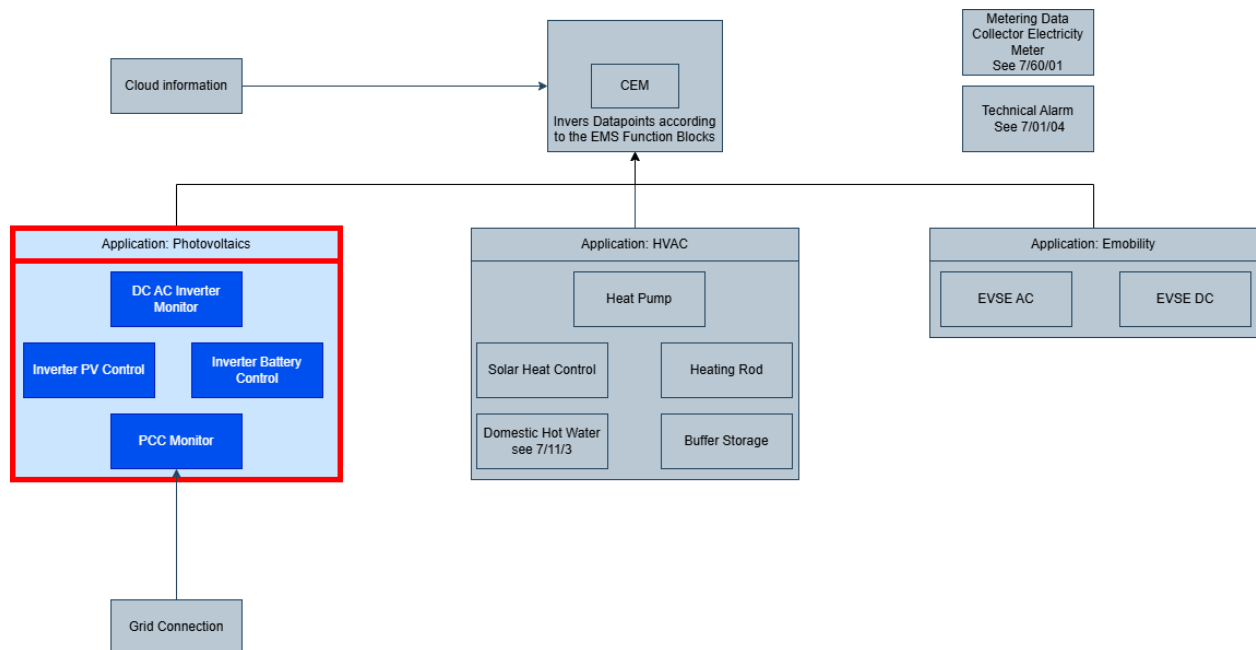


Figure 1 - Application “Photovoltaics”

2 FB Inverter PV Control

2.1 Aims and objectives

Photovoltaic (PV) panels turn direct solar radiation into direct current (DC) electrical power.

Characteristic capabilities of a solar PV panel are described by the maximum DC power, open circuit voltage, and short circuit current.

The conversion rate of solar radiation power to DC electrical power determines the efficiency of the solar PV panel.

A single solar cell only provides a low power output. For this reason, multiple solar cells are connected in series or in parallel to form a module (i.e. busbars).

A standard module, for example, consists of 60 cells. Typically, modules have an output of several hundred Wp (watt peak).

By connecting several solar modules, the output and thus the amount of solar power generated can be increased accordingly. The connection can be made in series or parallel. In series connection, the modules are always connected positive pole to negative pole one behind the other. The voltage of the individual modules adds up to the total voltage. The series-connected modules are referred to as a string. The negative and positive poles of the string are then connected to the (string) inverter.

In the parallel connection, a connection is made between the negative and positive poles of the individual modules and the (central) inverter. The voltage achieved corresponds to the voltage of a single photovoltaic module, but the currents are added together.

Photovoltaic components always generate direct current. As most electrical appliances in the residential environment run on alternating current for converting direct current into alternating current an inverter is used. The inverter feeds the DC electrical energy into the AC grid (e.g. 230 V AC).

The selection of inverter type and size is depending on the size of the system and the type of connection.

For autonomy the PV system may be combined with an electrical energy storage typically on the DC side. Alternatively, a power storage unit integrated on the AC side can be used.

To reduce the effects of shading or varying power, modern inverters use what is known as MPP tracking; MPP stands for Maximum Power Point. The task of the MPP tracker is to always retrieve the optimum power from the module or string depending on the solar radiation and temperature. Each inverter has an MPP voltage range within which it can operate MPP tracking.

The FB Inverter PV Control provides information about the maximum available and the currently provided power of a solar PV panel. Typically this control FB is located in the inverter device.

2.2 Functional Block diagram

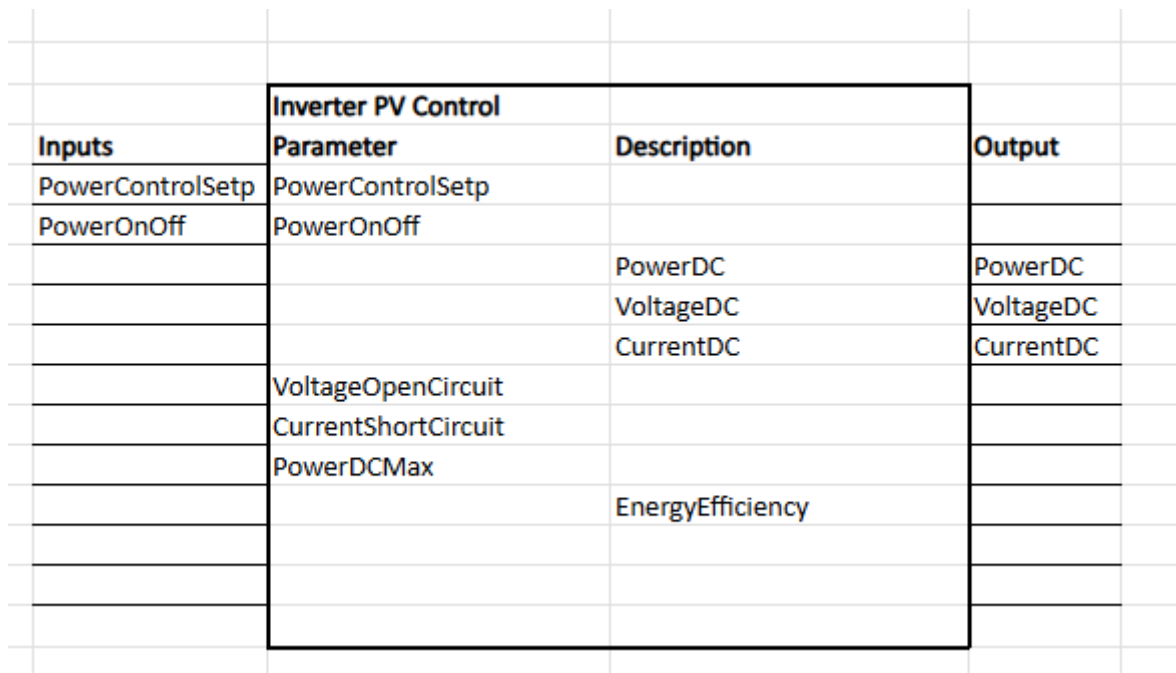


Figure 2 – Functional Block diagram Inverter PV Control

2.3 List of Datapoints – Inverter PV Control

Inputs		
Datapoint	Description/Remarks	Datapoint Type
PowerControlSetp	Setpoint for limiting the AC power output.	5.001 DPT_Scaling
PowerOnOff	Switch the AC power output of the inverter.	1.001 DPT_Switch

Outputs		
Datapoint	Description/Remarks	Datapoint Type
PowerDC	Present DC power of all solar panels connected to this PV control.	14.056 DPT_Value_Power
VoltageDC	Present DC voltage of all solar panels connected to this PV control.	14.027 DPT_Value_Electric_Potential
CurrentDC	Present DC current of all solar panels connected to this PV control.	14.019 DPT_Value_Electric_Current

Parameters		
Datapoint	Description/Remarks	Datapoint Type
VoltageOpenCircuit	Open circuit DC voltage of all solar panels connected to this PV control.	14.027 DPT_Value_Electric_Potential
CurrentShortCircuit	Short circuit DC current of all solar panels connected to this PV control.	14.019 DPT_Value_Electric_Current
PowerDCMax	Maximum power of all solar panels connected to this PV control (peak power).	14.056 DPT_Value_Power

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
EnergyEfficiency	Energy efficiency of the solar panels (produced peak DC power/ radiated power) calculated by the manufacturer.	5.001 DPT_Scaling

2.4 Detailed specification of the Datapoints

2.4.1 Inputs: Inverter PV Control

2.4.1.1 PowerControlSetp

DP Name:	PowerControlSetp	Abbr.:	-	Mandatory	☐		
FB Name:	Inverter PV Control			Can be internal	☐		
Description							
This is a percent value that shall limit the AC Power output of the PV. The Power on the AC side shall be reduced in relation to the Maximum DC Power (peak power).							
EXAMPLE 01 A value of 70% limits the Output of the AC Power to 70% of the Maximum DC Power.							
Datapoint Type							
DPT Name:	DPT_Scaling						
DPT Format:	U ₈	DPT ID:	5.001				
Field	Description	Supp.	Range	Unit	Default		
UnsignedValue	The percentage value of the power limitation.	M	full	%	none		
Access Type							
◆ Input							
N → this	☒	1 → this	☐				
Spontaneous	☒	Cyclically:	☐	Time-out:			
Request	☐	Polling:	☐	Period:			
Communication Type							
◆ Group Object Datapoint				Mandatory:	☒		
Default Group Address:		none					
◆ Property				Mandatory:	☒		
DP Address: (in the server)	IO Type(ID):	1250	Property ID:	51			
	Start-Index:	1	N° of elements:	1			
Property access:	Read only	☐	Read/Write	☒			
Protection *)	Read level	-	Write level	-			
Dynamics							
Power down:	Save:	☐					
Power up:	Value:	No initialisation:	☒	Default value:	☐		
		Saved value:	☐	Current value (not for input):	☐		
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐		
Exception Handling							
None.							
Special Features							
None.							

2.4.1.2 PowerOnOff

DP Name:	PowerOnOff	Abbr.:		Mandatory	☒
FB Name:	Inverter PV Control			Can be internal	☐
Description					
This is an on/off signal to switch the AC Power output of the PV On or Off. "On" means no limitation of the AC Power output, "Off" means no AC Power output.					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT Format:	B ₁	DPT_ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	This shall switch the AC power output.	M	full	none	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1250	Property ID:	52	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.4.2 Outputs: Inverter PV Control

2.4.2.1 PowerDC

DP Name:	PowerDC		Abbr.:		Mandatory	☐
FB Name:	Inverter PV Control				Can be internal	☐
Description						
Present DC power of all solar panels connected to this PV control						
Datapoint Type						
DPT_Name:	DPT_Value_Power					
DPT Format:	F ₃₂			DPT_ID:	14.056	
Field	Description	Supp.	Range	Unit	Default	
F ₃₂	DC power	M	full	W	none	
Access Type						
◆ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☒	Δ-Value:	> 1 %	
		Cyclic	☐	Period:		
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1250	Property ID:	60		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐		
Exception Handling						
None.						
Special Features						
None.						

2.4.2.2 VoltageDC

DP Name:	VoltageDC	Abbr.:		Mandatory	☐
FB Name:	Inverter PV Control			Can be internal	☐
Description					
Present DC voltage of all solar panels connected to this PV control.					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Potential				
DPT Format:	F ₃₂	DPT_ID:	14.027		
Field	Description	Supp.	Range	Unit	Default
F ₃₂	DC voltage	M	full	V	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	> 1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1250	Property ID:	61	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.4.2.3 CurrentDC

DP Name:	CurrentDC	Abbr.:		Mandatory	☐
FB Name:	Inverter PV Control			Can be internal	☐
Description					
Present DC current of the solar panels connected to this PV control					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Current				
DPT Format:	F ₃₂	DPT_ID:	14.019		
Field	Description	Supp.	Range	Unit	Default
F ₃₂	DC current	M	full	A	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1250	Property ID:	62	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
None.					
Special Features					
None.					

2.4.3 Parameters: Inverter PV Control

2.4.3.1 VoltageOpenCircuit

FB:		Property Name (Server):	VoltageOpenCircuit	Mandatory	☐
				Optional	☒
Description:					
Open circuit DC voltage of the solar panels connected to this PV control					
DPT:	Name	DPT_Value_Electric_Potential	DPT_ID	14.027	Datatype format: F ₃₂
Field	Description	Sup.	Range	Unit	Resol.: Default
			full	V	none
Communication:					
DP Address: (in the server)	Object Type:	1250	Property ID:	111	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection	Read level	-	Write level	-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
None.					
Special Features:					
None.					

2.4.3.2 CurrentShortCircuit

FB:		Property Name (Server):	CurrentShortCircuit			Mandatory	☐	Optional	☒
Description:									
Short circuit DC current of all solar panels connected to this PV control.									
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:	F ₃₂			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
				full	A		none		
Communication:									
DP Address: (in the server)		Object Type:	1250	Property ID:		112			
		Start-Index:	1	N° of elements:		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
None.									
Special Features:									
None.									

2.4.3.3 PowerDCMax

FB:		Property Name (Server):	PowerDCMax			Mandatory	☐	Optional	☒
Description:									
Maximum power of the solar panels connected to this PV control (peak power). In an installation this object is normally acting on the AC side of the PV systems i.e. inverter. Inverter losses are neglectable for this function and hence for structural reasons this object is placed within the PV functional block i.e. on the DC side									
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:	F ₃₂			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
				full	W		none		
Communication:									
DP Address: (in the server)		Object Type:	1250	Property ID:		113			
		Start-Index:	1	N° of elements:		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
None.									
Special Features:									
None.									

2.4.4 Diagnostic Data: Inverter PV Control

2.4.4.1 EnergyEfficiency

FB:		Property Name (Server):	EnergyEfficiency		Mandatory	☐
					Optional	☒
Description:						
Energy efficiency of the solar panels according EN 50530 as given by the manufacturer						
DPT:	Name		DPT_ID	5.001	Datatype format:	
Field	Description		Sup.	Range	Unit	Resol.: Default
				full	%	none
Communication:						
DP Address:		Object Type:		1250	Property ID:	
(in the server)		Start-Index:		1	N° of elements:	
Property access:		Read only		☒ -	Read/Write	
Protection		Read level		-	Write level	
					-	
Exception Handling:						
Value after Power-up:		Stored Value		☒	Curr Value	☐
					Default Value	☐
Special Features:						
None.						

3 FB Inverter Battery Control

3.1 Aims and objectives

A battery control system monitors, controls, and secures every single battery cell to guarantee the best lifespan, availability, and performance of the battery stack. To fulfil this task the battery control system measures different parameters like voltage (current, temperature) of every single battery cell as well as the battery stack environment temperature.

Depending on these measured values and the battery chemistry the battery controller sets limits for charging/discharging, calculates the energy flexibility (how much energy could be provided, how much energy could be stored) and the losses (self-discharging) of the battery stack.

This data for charging/discharging, energy flexibility and losses shall be communicated to the Energy management system.

The main battery states of charge (SoC) are shown in the figure below:

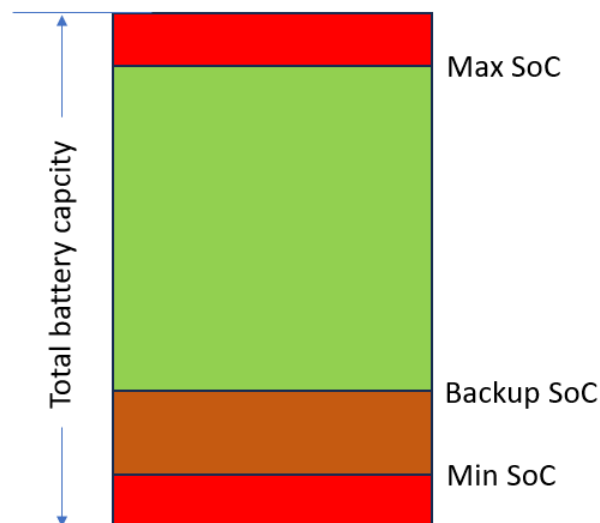


Figure 3 – Main Battery States of Charge

This functional block represents a battery storage and provides the objects for its control and for monitoring the states and values.

Typically, this functional block is located in the inverter device.

3.2 Functional Block Diagram: Inverter Battery Control

Inverter Battery Control			
Inputs	Parameter	Description	Output
ForceGridCharging	ForceGridCharging		
DisableGridDischarging	DisableGridDischarging		
CapacityBackup	CapacityBackup		
		BatteryOpMode	BatteryOpMode
		StateOfCharge	StateOfCharge
		StateOfHealth	StateOfHealth
		CapacityEnergyEff	CapacityEnergyEff
		PowerEff	PowerEff
		PowerChargeMax	PowerChargeMax
		PowerChargeMin	PowerChargeMin
		DepthOfDischarge	DepthOfDischarge
		Fault	Fault
		LoadCyclesCount	LoadCyclesCount
		TempStorage	TempStorage
		VoltageDC	VoltageDC
	StateOfChargeMin		
	StateOfChargeMax		
		StateOfChargeReserveMin	
		StateOfChargeReserveMax	
		EnergyMax	
		EnergyEfficiency	

Figure 4 – Functional Block diagram Inverter Battery Control

3.3 List of Datapoints: Inverter Battery Control

Inputs		
Datapoint	Description/Remarks	Datapoint Type
ForceGridCharging	Command for forcing the charging of the battery. This allows to override the principle of storing self-produced power to source low-cost energy from the grid.	1.002 DPT_Bool
DisableGridDischarging	Command for inhibiting discharging of the battery. This allows to maintain the current load in case of grid power failure situations or for lowering the energy cost from the grid.	1.002 DPT_Bool
CapacityBackup	Energy that should always remain in the battery for back-up purposes. This value is controlled by the energy management system and is limited between StateOfChargeReserveMin and StateOfChargeReserveMax.	5.001 DPT_Scaling

Outputs		
Datapoint	Description/Remarks	Datapoint Type
BatteryOpMode	Monitors the mode of the battery. States are reflect whether the battery is decoupled (not connected), is initializing or connected.	20.1218 DPT_OP_Mode_Bat
StateOfCharge	State of Charge shows the present battery charge level where 100 % is maximum State of Charge (SoCMax) and 0% is minimum State of Charge (SoCmin)	5.001 DPT_Scaling
StateOfHealth	Monitors the ratio between maximum capacity of the battery at the moment of production and the present maximum capacity of the battery	5.001 DPT_Scaling
CapacityEnergyEff	Present energy capacity available in the battery Energy_Present_Cap = SoCmax – SoCmin – SoCbackup	13.010 DPT_ActiveEnergy
PowerEff	Present DC Power charged or discharged to or from the battery	14.056 DPT_Value_Power
PowerChargeMax	Present maximum possible DC power for charging the battery according to the battery charging curve.	14.056 DPT_Value_Power
PowerChargeMin	Present maximum possible DC power for discharging the battery according to the battery discharging curve.	14.056 DPT_Value_Power
DepthOfDischarge	Depth of Discharge (DoD) shows the present battery discharge level where 0 % is maximum State of Charge (StateOfChargeMax) and 100% is minimum State of Charge (StateOfChargeMin). The sum of SoC and DoD is 100 %	5.001 DPT_Scaling
Fault	Global error message of the battery where true means "There is an error" and false means "There is no error". For detailed error or alarm messages the Functional Block "technical alarm" shall be used, see KNX Standard 7/1/4 Technical Alarm	1.005 DPT_Alarm
LoadCyclesCount	Number of complete charging cycles since installation of the battery.	12.001 DPT_Value_4_Ucount
TempStorage	Temperature of the cells in the battery	9.001 DPT_Value_Temp
VoltageDC	DC Voltage at the connection points of the battery	14.027 DPT_Value_Electric_Potential

Parameters		
Datapoint	Description/Remarks	Datapoint Type
StateOfChargeMin	To prevent reduction of the battery lifetime, the full capacity range is not used. A small capacity of the battery remains unused. This level is the Minimum State of Charge. Typically, the value is about 10% and is normally stated in the data sheet of the battery.	5.001 DPT_Scaling
StateOfChargeMax	To prevent reduction of the battery lifetime, the full capacity range is not used. A small capacity of the battery remains unused. The maximum used level is called the maximum state of Charge. Typically, the value is about 90% and is normally stated in the data sheet of the battery.	5.001 DPT_Scaling

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
StateOfChargeReserveMin	This is the lower limit for the backup capacity functionality, where 0 % is StateOfChargeMin and 100% is StateOfChargeMax. This value has to be lower than the upper limit. Value typically found in the data sheet of the battery.	5.001 DPT_Scaling
StateOfChargeReserveMax	This is the upper limit for the backup capacity functionality, where 0 % is StateOfChargeMin and 100% is StateOfChargeMax. This value has to be higher than the upper limit.	5.001 DPT_Scaling
EnergyMax	Maximum capacity of the new battery according to the data sheet.	13.010 DPT_ActiveEnergy
EnergyEfficiency	This is the ratio between energy for discharging and the energy for charging the battery.	5.001 DPT_Scaling

3.4 Detailed specification of the Datapoints

3.4.1 Inputs: Inverter Battery Control

3.4.1.1 ForceGridCharging

DP Name:	ForceGridCharging	Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Command for forcing the charging of the battery. This allows to override the principle of storing self-produced power to source low-cost energy from the grid.					
Datapoint Type					
DPT_Name:	DPT_Bool				
DPT Format:	B ₁	DPT_ID:	1.002		
Field		Supp.	Range	Unit	Default
b	With this the charging of the battery can be forced (1 = on) or again deactivated (0 = off)	M	full	bool	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous Request	☒	Cyclically:	☐	Time-out:	
	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID): Start-Index:	1251 1	Property ID: N° of elements:	51 1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

3.4.1.2 DisableGridDischarging

DP Name:	DisableGridDischarging	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Command for inhibiting discharging of the battery. This allows to maintain the current load in case of grid power failure situations or for lowering the energy cost from the grid.					
Datapoint Type					
DPT_Name:	DPT_Bool				
DPT Format:	B ₁	DPT_ID:	1.002		
Field		Supp.	Range	Unit	Default
b	With this the charging of the battery can be inhibited (1 = on) or again deactivated (0 = off)	M	full	bool	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	52	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

3.4.1.3 CapacityBackup

DP Name:	CapacityBackup	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Energy that should always remain in the battery for back-up purposes. This value is controlled by the energy management system and is limited between StateOfChargeReserveMin and StateOfChargeReserveMax.					
Datapoint Type					
DPT Name:	DPT_Scaling				
DPT Format:	U ₈	DPT ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Setting of the backup capacity of the battery	M	full	%	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	53
		Start-Index:	1	N° of elements:	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling					
Values below StateOfChargeReserveMin are limited to StateOfChargeReserveMin, values above StateOfChargeReserveMax are limited to StateOfChargeReserveMax					
Special Features					

3.4.2 Outputs: Inverter Battery Control

3.4.2.1 BatteryOpMode

DP Name:	BatteryOpMode	Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Monitors the mode of the battery. States are reflect whether the battery is decoupled (not connected), is initializing or connected.					
Datapoint Type					
DPT Name:	DPT_OP_Mode_Bat				
DPT Format:	N ₈	DPT ID:	20.1218		
Field		Supp.	Range	Unit	Default
BatteryOp Mode	0 = decoupled 1 = initializing 2 = connected 3 to 255 = not used, reserved	M	full	-	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	60	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

3.4.2.2 StateOfCharge (SoC)

DP Name:	StateOfCharge (SoC)		Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control				Can be internal	☐
Description						
State of Charge shows the present battery charge level where 100 % is maximum State of Charge (SoCMax) and 0% is minimum State of Charge (SoCmin)						
Datapoint Type						
DPT_Name:	DPT_Scaling					
DPT Format:	U ₈		DPT_ID:	5.001		
Field			Supp.	Range	Unit	Default
Unsigned Value	Percentage reflecting the present battery charge level		M	full	%	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	61	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

3.4.2.3 StateOfHealth

DP Name:	StateOfHealth		Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control				Can be internal	☐
Description						
Monitors the ratio between maximum capacity of the battery at the moment of production and the present maximum capacity of the battery						
Datapoint Type						
DPT_Name:	DPT_Scaling					
DPT Format:	U ₈		DPT_ID:	5.001		
Field			Supp.	Range	Unit	Default
Unsigned Value	Percentage reflecting the present state of health of the battery		M	full	%	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	62	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

3.4.2.4 CapacityEnergyEff

DP Name:	CapacityEnergyEff		Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control				Can be internal	☐
Description						
Present energy capacity available in the battery Energy_Present_Cap = SoCmax – SoCmin – SoCbackup						
Datapoint Type						
DPT_Name:	DPT_ActiveEnergy					
DPT Format:	V ₃₂		DPT_ID:	13.010		
Field			Supp.	Range	Unit	Default
SignedValue	Value reflecting the present energy capacity available in the battery		M	full	Wh	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	1251				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	63	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

3.4.2.5 PowerEff

DP Name:	PowerEff		Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control				Can be internal	☐
Description						
Present DC Power (charged or discharged) to or from the battery When the battery is charged the value is positive (+). When the battery is discharged the value is negative (-).						
Datapoint Type						
DPT Name:	DPT_Value_Power					
DPT Format:	F ₁₆	DPT ID:	14.056			
Field		Supp.	Range	Unit	Default	
Float/Value	Value reflecting DC power (dis)charged	M	full	W	none	
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	64	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

3.4.2.6 PowerChargeMax

DP Name:	PowerChargeMax	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Present maximum possible DC power for charging the battery according to the battery charging curve.					
Datapoint Type					
DPT_Name:	DPT_Value_Power				
DPT Format:	F ₁₆	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
FloatValue	Value reflecting maximum possible DC charging power	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	65	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

3.4.2.7 PowerChargeMin

DP Name:	PowerChargeMin	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Present maximum possible DC power for discharging the battery according to the battery discharging curve.					
Datapoint Type					
DPT_Name:	DPT_Value_Power				
DPT Format:	F ₁₆	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
FloatValue	Value reflecting minimum possible DC discharging power		full	W	none
Access Type					
◆ Output					
	this → M	☒	this → 1	☐	
	Spontaneous	☒	COV:	☒	Δ-Value: >1 %
			Cyclic	☐	Period:
	Request	☒			
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	66
		Start-Index:	1	N° of elements:	1
	Property access:	Read only	☒	Read/Write	☐
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

3.4.2.8 DepthOfDischarge (DoD)

DP Name:	DepthOfDischarge (DoD)	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Depth of Discharge shows the present battery discharge level where 0 % is maximum State of Charge (StateOfChargeMax) and 100% is minimum State of Charge (StateOfChargeMin). The sum of SoC and DoD is 100 %					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value reflecting the present battery discharge level	M	full	%	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	67	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

3.4.2.9 Fault

DP Name:	Fault		Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐	
Description						
Global error message of the battery where TRUE means "There is an error" and FALSE means "There is no error". For detailed error or alarm messages the Functional Block "technical alarm" shall be used, see KNX Standard 7/1/4 Technical Alarm						
Datapoint Type						
DPT_Name:	DPT_Alarm					
DPT Format:	B ₁	DPT_ID:	1.005			
Field		Supp.	Range	Unit	Default	
b	Status whereby 1 = error and 0 = no error	M	full	bool	none	
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1251	Property ID:	68	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

3.4.2.10 LoadCyclesCount

DP Name:	LoadCyclesCount	Abbr.:		Mandatory	☒
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Number of complete charging cycles since installation of the battery.					
Datapoint Type					
DPT Name:	DPT Value 4 Ucount				
DPT Format:		DPT ID:	12.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value reflecting number of complete charging cycles since installation	M	full	-	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	69	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

3.4.2.11 TempStorage

DP Name:	TempStorage	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
Temperature of the cells in the battery					
Datapoint Type					
DPT_Name:	DPT_Value_Temp				
DPT Format:		DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting battery cells temperature	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	70	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

3.4.2.12 VoltageDC

DP Name:	VoltageDC	Abbr.:		Mandatory	☐
FB Name:	Inverter Battery Control			Can be internal	☐
Description					
DC Voltage at the connection points of the battery					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Potential				
DPT_Format:		DPT_ID:	14.027		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting DC voltage at battery connection points	M	full	V	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1251	Property ID:	71	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

3.4.3 Parameter: Inverter Battery Control

3.4.3.1 StateOfChargeMin

FB:	Inverter Battery Control	Property Name (Server):	StateOfChargeMin		Mandatory	☐	Optional	☒
Description:								
To prevent reduction of the battery lifetime, the full capacity range is not used. A small capacity of the battery remains unused. This level is the Minimum State of Charge. Typically, the value is about 10% and is normally stated in the data sheet of the battery.								
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Unsigned Value	Parameter determining the minimum state of charge		M	0 to 50	%	0,4 %	none	
Communication:								
DP Address: (in the server)		Object Type: 1251		Property ID: 111		Start-Index: 1 N° of elements: 1		
Property access:		Read only ☐ - Read/Write ☒						
Protection		Read level -		Write level -				
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐								
Special Features:								
None.								

3.4.3.2 StateOfChargeMax

FB:	Inverter Battery Control	Property Name (Server):	StateOfChargeMax		Mandatory	☐	Optional	☒
Description:								
To prevent reduction of the battery lifetime, the full capacity range is not used. A small capacity of the battery remains unused. The maximum used level is called the maximum state of Charge. Typically, the value is about 90% and is normally stated in the data sheet of the battery.								
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Unsigned Value	Parameter determining the maximum state of charge		M	50 to 100	%	0,4%	none	
Communication:								
DP Address: (in the server)		Object Type: 1251		Property ID: 112		Start-Index: 1 N° of elements: 1		
Property access:		Read only ☐ - Read/Write ☒						
Protection		Read level -		Write level -				
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐								
Special Features:								
None.								

3.4.4 Diagnostic Data: Inverter Battery Control

3.4.4.1 StateOfChargeReserveMin

FB:	Inverter Battery Control	Property Name (Server):	StateOfChargeReserveMin	Mandatory	☐	Optional	☒
Description:							
This is the lower limit for the backup capacity functionality, where 0 % is StateOfChargeMin and 100% is StateOfChargeMax. This value has to be lower than the upper limit. Value typically found in the data sheet of the battery.							
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Unsigned Value	Data indicating the lower limit for the backup capacity		M	full	%	0,4%	none
Communication:							
DP Address: (in the server)		Object Type: 1251	Property ID: 120		N° of elements: 1		
Property access:		Read only ☒ -	Read/Write ☐				
Protection		Read level -	Write level -				
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

3.4.4.2 StateOfChargeReserveMax

FB:	Inverter Battery Control	Property Name (Server):	StateOfChargeReserveMax	Mandatory	☐	Optional	☒
Description:							
This is the upper limit for the backup capacity functionality, where 0 % is StateOfChargeMin and 100% is StateOfChargeMax. This value has to be higher than the upper limit.							
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Unsigned Value	Data indicating the upper limit for the backup capacity		M	full	%	0,4%	none
Communication:							
DP Address: (in the server)		Object Type: 1251	Property ID: 121		N° of elements: 1		
Property access:		Read only ☒ -	Read/Write ☐				
Protection		Read level -	Write level -				
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

3.4.4.3 EnergyMax

FB:	Inverter Battery Control	Property Name (Server):	EnergyMax			Mandatory	☒	
						Optional	☐	
Description:								
Maximum capacity of the new battery according to the data sheet.								
DPT:	Name	DPT_ActiveEnergy	DPT_ID	13.010	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Signed Value	Data informing about the max. capacity of the new battery		M	full	Wh	1 Wh	none	
Communication:								
DP Address: (in the server)		Object Type:	1251	Property ID:	122			
		Start-Index:	1	N° of elements:	1			
Property access:		Read only	☒ -	Read/Write	☐			
Protection		Read level	-	Write level	-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Special Features:								
None.								

3.4.4.4 EnergyEfficiency

FB:	Inverter Battery Control	Property Name (Server):	EnergyEfficiency			Mandatory	☐	
						Optional	☒	
Description:								
This is the ratio between energy for discharging and the energy for charging the battery								
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Unsign ed Value	Ratio between energy for charging and discharging		M	full	%	0,4%	none	
Communication:								
DP Address: (in the server)		Object Type:	1251	Property ID:	123			
		Start-Index:	1	N° of elements:	1			
Property access:		Read only	☒ -	Read/Write	☐			
Protection		Read level	-	Write level	-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Special Features:								
None.								

3.4.5 New Data Point Types

3.4.5.1 Operation Mode Battery

<u>Format:</u>	1 octet: N ₈			
octet nr.	1			
field names	field1			
encoding	NNNNNNNN			
<u>Encoding:</u>	Encoding absolute value N = [0 ... 255]			
<u>Unit:</u>	none			
<u>Resol.:</u>	none			
<u>PDT:</u>	PDT_ENUM8 (alt: PDT_UNSIGNED_CHAR)			
Datapoint Types				
<u>ID:</u>	<u>Name:</u>	<u>Encoding:</u>	<u>Range:</u>	<u>Use:</u>
20.1218	DPT_OP_Mode_Bat	Monitors the mode of the battery. States are 0 = decoupled 1 = initializing 2 = connected 3 to 255 = not used; reserved	[0 to 2]	FB

4 FB DC-AC Inverter Monitor

4.1 Aims and objectives

An inverter is an electronic component that converts electrical current.

PV systems and battery systems are by nature using DC to feed the generated current into the AC grid.

Typically in an optimised EMS self-consumption of the generated electricity may have priority. The pathway of the current is then controlled by the inverter, which is controlled by an EMS management device.

This functional block addresses the control part of the inverter while the generation and storage parts are addressed in separate functional blocks i.e., Inverter PV control and Inverter Battery Control.

This Functional Block serves to control and monitor the values.

4.2 Functional Block Diagram: DC-AC Inverter Monitor

Inputs	DC AC Inverter Monitor		Output
	Parameter	Description	
			CurrentDC
			PowerDC
			EnergyEfficiency
			VoltageL1L2
			VoltageL2L3
			VoltageL3L1
			VoltageL1
			VoltageL2
			VoltageL3
			CurrentL1
			CurrentL2
			CurrentL3
			CurrentTotal
			MainsFrequency
			ActivePowerTotal
			ApparentPowerTotal
			ReactivePowerTotal
			PowerFactor
			ActiveEnergySummation
			TempInverter

Figure 5 – DC-AC Inverter Monitor

4.3 List of Datapoints: DC-AC Inverter Monitor

Outputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentDC	Present DC Current value (DCA) of the inverter input	14.019 DPT_Value_Electric_Current
PowerDC	Present DC power value (DCW) of the inverter input	14.056 DPT_Value_Power
VoltageL1L2	AC voltage of the output between phase L1 and L2	14.027 DPT_Value_Electric_Potential
VoltageL2L3	AC voltage of the output between phase L2 and L3	14.027 DPT_Value_Electric_Potential
VoltageL3L1	AC voltage of the output between phase L3 and L1	14.027 DPT_Value_Electric_Potential
VoltageL1	AC voltage of the output between phase L1 and N	14.027 DPT_Value_Electric_Potential
VoltageL2	AC voltage of the output between phase L2 and N	14.027 DPT_Value_Electric_Potential
VoltageL3	AC voltage of the output between phase L3 and N	14.027 DPT_Value_Electric_Potential
CurrentL1	AC current of the output on phase L1	14.019 DPT_Value_Electric_Current
CurrentL2	AC current of the output on phase L2	14.019 DPT_Value_Electric_Current
CurrentL3	AC current of the output on phase L3	14.019 DPT_Value_Electric_Current
CurrentTotal	AC total current value, sum of all phases	14.019 DPT_Value_Electric_Current
MainsFrequency	Present frequency value (Hz) of the output	14.033 DPT_Value_Frequency
ActivePowerTotal	AC Power value sum of all phases	14.056 DPT_Value_Power
ApparentPowerTotal	Apparent Power (VA)	14.080 DPT_Value_ApparentPower
ReactivePowerTotal	Reactive Power (VAr)	14.081 DPT_Value_ReactivePower
PowerFactor	Power Factor (PF)	14.057 DPT_Value_Power_Factor
ActiveEnergySummation	AC Lifetime Energy production	13.010 DPT_ActiveEnergy
TempInverter	Present temperature of the inverter	9.001 DPT_Value_Temp

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
EnergyEfficiency	Energy efficiency of conversion from DC power to AC power. [Energy efficiency = Present AC Power/Present DC Power (in %)] When there is no conversion, the value is 0 (zero).	5.001 DPT_Scaling

4.4 Detailed specification of the Datapoints

4.4.1 Inputs: DC-AC Inverter Monitor

This Functional Block does not have inputs

4.4.2 Outputs: DC-AC Inverter Monitor

4.4.2.1 CurrentDC

DP Name:	CurrentDC			Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor				Can be internal	☐	
Description							
Present DC current value (DCA) of the inverter input							
Datapoint Type							
DPT_Name:							
DPT Format:	DPT_Value_Electric_Current			DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default		
Float Value	Value reflecting present DC current value of inverter input			M	full	A	none
Access Type							
◆ Output							
this → M	☒	this → 1	☐				
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %		Min repetition time:
		Cyclic	☐	Period:			
Request	☒						
Communication Type							
◆ Group Object Datapoint						Mandatory:	☒
Default Group Address:		None					
◆ Property						Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	51			
	Start-Index:	1	N° of elements:	1			
Property access:	Read only	☒	Read/Write	☐			
Protection *)	Read level	-	Write level	-			
Dynamics							
Power down:	Save:	☐					
Power up:	Value:	No initialisation:	☒	Default value:	☐		
		Saved value:	☐	Current value (not for input):	☐		
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐		
Exception Handling							
Special Features							

4.4.2.2 PowerDC

DP Name:	PowerDC	Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present DC Power value (DCW) of the inverter input					
Datapoint Type					
DPT_Name:	DPT_Value_Power				
DPT Format:		DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting DC Power value of inverter input	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	52	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.2.3 VoltageL1L2

DP Name:	VoltageL1L2	Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present AC Voltage of the output between phase L1 and L2					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Potential				
DPT_Format:		DPT_ID:	14.027		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting AC Voltage of the output L1/L2	M	full	V	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	53	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.2.4 VoltageL2L3

DP Name:	VoltageL2L3	Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present AC Voltage of the output between phase L2 and L3					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Potential				
DPT_Format:		DPT_ID:	14.027		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting AC Voltage of the output L2/L3	M	full	V	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	54	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.2.5 VoltageL3L1

DP Name:	VoltageL3L1			Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor					Can be internal	☐
Description							
Present AC Voltage of the output between phase L3 and L1							
Datapoint Type							
DPT Name:	DPT_Value_Electric_Potential						
DPT Format:		DPT ID:	14.027				
Field		Supp.	Range	Unit	Default		
Float Value	Value reflecting AC Voltage of the output L3/L1		M	full	V	none	
Access Type							
◆ Output							
	this → M	☒	this → 1	☐			
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %	
			Cyclic	☐	Period:		
	Request	☒					
Communication Type							
◆ Group Object Datapoint						Mandatory:	☒
	Default Group Address:	none					
◆ Property						Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1252	Property ID:	55		
		Start-Index:	1	N° of elements:	1		
	Property access:	Read only	☒	Read/Write	☐		
	Protection *)	Read level	-	Write level	-		
Dynamics							
	Power down:	Save:	☐				
	Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐		
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐		
Exception Handling							
Special Features							

4.4.2.6 VoltageL1

DP Name:	VoltageL1	Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present AC Voltage of the output between phase L1 and N					
Datapoint Type					
DPT_Name:					
DPT Format:		DPT_ID:	14.027		
Field		Supp.	Range	Unit	Default
Float Value	Value reflecting AC Voltage of the output L1/N	M	full	V	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	56	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.2.7 VoltageL2

DP Name:	VoltageL2		Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor				Can be internal	☐
Description						
Present AC Voltage of the output between phase L2 and N						
Datapoint Type						
DPT Name:						
DPT Format:				DPT ID:	14.027	
Field				Supp.	Range	Unit
Float Value	Value reflecting AC Voltage of the output L2/N			M	full	V
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1252	Property ID:	57	
		Start-Index:	1	N° of elements:	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

4.4.2.8 VoltageL3

DP Name:	VoltageL3		Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor				Can be internal	☐
Description						
Present AC Voltage of the output between phase L3 and N						
Datapoint Type						
DPT Name:						
DPT Format:				DPT ID:	14.027	
Field				Supp.	Range	Unit
Float Value	Value reflecting AC Voltage of the output L3/N			M	full	V
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	58		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

4.4.2.9 CurrentL1

DP Name:	CurrentL1			Abbr.:		Mandatory	☐																								
FB Name:	DC/AC Inverter Monitor					Can be internal	☐																								
Description																															
Present AC Current of the output on phase L1																															
Datapoint Type																															
DPT Name:	DPT_Value_Electric_Current																														
DPT Format:		DPT_ID:	14.019																												
Field		Supp.	Range	Unit	Default																										
Float Value	Value reflecting present AC current output L1		M	full	A	none																									
Access Type																															
◆ Output																															
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒									
this → M	☒	this → 1	☐																												
Spontaneous	☒	COV:	☒																												
		Δ-Value:	>1 %																												
		Cyclic	☐																												
		Period:																													
Request	☒																														
Communication Type																															
◆ Group Object Datapoint						Mandatory:	☒																								
Default Group Address:		none																													
◆ Property						Mandatory:	☒																								
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	59																											
	Start-Index:	1	N° of elements:	1																											
Property access:	Read only	☒	Read/Write	☐																											
Protection *)	Read level	-	Write level	-																											
Dynamics																															
Power down:	Save:	☐																													
Power up:	Value:	No initialisation:	☒	Default value:		☐																									
		Saved value:	☐	Current value (not for input):		☐																									
	Transmit on bus (only for output):		☐	Read from bus (only for input):		☐																									
Exception Handling																															
Special Features																															

4.4.2.10 CurrentL2

DP Name:	CurrentL2					Mandatory	☐																
FB Name:	DC/AC Inverter Monitor					Can be internal	☐																
Description																							
Present AC Current of the output on phase L2																							
Datapoint Type																							
DPT_Name:	DPT_Value_Electric_Current																						
DPT_Format:				DPT_ID:	14.019																		
Field				Supp.	Range	Unit	Default																
Float Value	Value reflecting present AC current output L2			M	full	A	none																
Access Type																							
◆ Output																							
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Cyclic	☐	Request	☒			Δ-Value:		>1 %		Min repetition time:		
this → M	☒	this → 1	☐																				
Spontaneous	☒	COV:	☒																				
		Cyclic	☐																				
Request	☒																						
Communication Type																							
◆ Group Object Datapoint						Mandatory:	☒																
Default Group Address:		none																					
◆ Property						Mandatory:	☒																
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	60																			
	Start-Index:	1	N° of elements:	1																			
Property access:	Read only	☒	Read/Write	☐																			
Protection *)	Read level	-	Write level	-																			
Dynamics																							
Power down:	Save:	☐																					
Power up:	Value:	No initialisation:	☒	Default value:	☐																		
		Saved value:	☐	Current value (not for input):	☐																		
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																		
Exception Handling																							
Special Features																							

4.4.2.11 CurrentL3

DP Name:	CurrentL3	Abbr.:		Mandatory	☐																									
FB Name:	DC/AC Inverter Monitor			Can be internal	☐																									
Description																														
Present AC Current of the output on phase L3																														
Datapoint Type																														
DPT Name:	DPT_Value_Electric_Current																													
DPT Format:		DPT ID:	14.019																											
Field		Supp.	Range	Unit	Default																									
Float Value	Value reflecting present AC current output L3	M	full	A	none																									
Access Type																														
◆ Output																														
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td colspan="2"></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒								
this → M	☒	this → 1	☐																											
Spontaneous	☒	COV:	☒																											
		Δ-Value:	>1 %																											
		Cyclic	☐																											
		Period:																												
Request	☒																													
Communication Type																														
◆ Group Object Datapoint				Mandatory:	☒																									
Default Group Address:		none																												
◆ Property				Mandatory:	☒																									
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	61																										
	Start-Index:	1	N° of elements:	1																										
Property access:	Read only	☒	Read/Write	☐																										
Protection *)	Read level	-	Write level	-																										
Dynamics																														
Power down:	Save:	☐																												
Power up:	Value:	No initialisation:	☒	Default value:	☐																									
		Saved value:	☐	Current value (not for input):	☐																									
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																									
Exception Handling																														
Special Features																														

4.4.2.12 CurrentTotal

DP Name:	CurrentTotal			Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor					Can be internal	☐
Description							
Present AC Total Current value, sum of all phases							
Datapoint Type							
DPT Name:	DPT_Value_Electric_Current						
DPT Format:		DPT ID:	14.019				
Field		Supp.	Range	Unit	Default		
Float Value	Value reflecting total current all phases		M	full	A	none	
Access Type							
◆ Output							
	this → M	☒	this → 1	☐			
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %	Min repetition time:
			Cyclic	☐	Period:		
	Request	☒					
Communication Type							
◆ Group Object Datapoint						Mandatory:	☒
	Default Group Address:	none					
◆ Property						Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1252	Property ID:	62		
		Start-Index:	1	N° of elements:	1		
	Property access:	Read only	☒	Read/Write	☐		
	Protection *)	Read level	-	Write level	-		
Dynamics							
	Power down:	Save:	☐				
	Power up:	Value:	No initialisation:	☒	Default value:	☐	
			Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling							
Special Features							

4.4.2.13 MainsFrequency

DP Name:	MainsFrequency	Abbr.:		Mandatory	☐																								
FB Name:	DC/AC Inverter Monitor			Can be internal	☐																								
Description																													
Present Frequency value of the voltage output																													
Datapoint Type																													
DPT Name:																													
DPT Format:	14.033 DPT_Value_Frequency	DPT ID:	14.033																										
Field		Supp.	Range	Unit	Default																								
Float Value	Value representing present frequency of output	M	full	Hz	none																								
Access Type																													
◆ Output																													
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒							
this → M	☒	this → 1	☐																										
Spontaneous	☒	COV:	☒																										
		Δ-Value:	>1 %																										
		Cyclic	☐																										
		Period:																											
Request	☒																												
Communication Type																													
◆ Group Object Datapoint				Mandatory:	☒																								
Default Group Address:		none																											
◆ Property				Mandatory:	☒																								
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	63																									
	Start-Index:	1	N° of elements:	1																									
Property access:	Read only	☒	Read/Write	☐																									
Protection *)	Read level	-	Write level	-																									
Dynamics																													
Power down:	Save:	☐																											
Power up:	Value:	No initialisation:	☒	Default value:	☐																								
		Saved value:	☐	Current value (not for input):	☐																								
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																								
Exception Handling																													
Special Features																													

4.4.2.14 ActivePowerTotal

DP Name:	ActivePowerTotal	Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present AC Power value as sum of all phases					
Datapoint Type					
DPT_Name:	14.056 DPT_Value_Power				
DPT Format:		DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float Value	Value representing present AC power sum all phases	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:	none				
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	64	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

4.4.2.15 ApparentPowerTotal

DP Name:	ApparentPowerTotal		Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor				Can be internal	☐
Description						
Present Apparent Power measured at the connection point to the AC power system						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_ApparentPower			DPT_ID:	14.080	
Field		Supp.	Range	Unit	Default	
Float Value	Value representing present apparent power			M	full	VA none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	65		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

4.4.2.16 ReactivePowerTotal

DP Name:	ReactivePowerTotal			Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor					Can be internal	☐
Description							
Present Reactive Power measured at the connection point to the AC power system							
Datapoint Type							
DPT Name:							
DPT Format:	DPT_Value_ReactivePower				DPT_ID:	14.081	
Field		Supp.	Range	Unit	Default		
Float Value	Value representing reactive power				M	full	VAR none
Access Type							
◆ Output							
	this → M	☒	this → 1	☐			
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %	Min repetition time:
			Cyclic	☐	Period:		
	Request	☒					
Communication Type							
◆ Group Object Datapoint						Mandatory:	☒
	Default Group Address:	none					
◆ Property						Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1252	Property ID:	66		
		Start-Index:	1	N° of elements:	1		
	Property access:	Read only	☒	Read/Write	☐		
	Protection *)	Read level	-	Write level	-		
Dynamics							
	Power down:	Save:	☐				
	Power up:	Value:	No initialisation:	☒	Default value:	☐	
			Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling							
Special Features							

4.4.2.17 PowerFactor

DP Name:	PowerFactor	Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present Power Factor (PF) measured at the connection point to the AC power system					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Value_Power_Factor	DPT_ID:	14.057		
Field		Supp.	Range	Unit	Default
Float Value	Value representing the present power factor	M	full	-	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	67	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

4.4.2.18 ActiveEnergySummation

DP Name:	ActiveEnergySummation	Abbr.:		Mandatory	☐
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Total AC active energy produced over the lifetime of the DC / AC inverter.					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_ActiveEnergy	DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Integer	Value representing total AC active energy produced	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	68	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.2.19 TempInverter

DP Name:	TempInverter	Abbr.:		Mandatory	☒
FB Name:	DC/AC Inverter Monitor			Can be internal	☐
Description					
Present temperature of the inverter					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float Value	Value representing current inverter temperature	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1252	Property ID:	69	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

4.4.3 Parameters: DC-AC Inverter Monitor

This Functional Block does not have parameters.

4.4.4 Diagnostic Data: DC-AC Inverter Monitor

4.4.4.1 Output: EnergyEfficiency

FB:	DC/AC Inverter Monitor	Property Name (Server):	EnergyEfficiency			Mandatory ☐ Optional ☒	
Description:							
Energy efficiency of conversion from DC power to AC power. [Energy efficiency = Present AC Power/Present DC Power (in %)]							
DPT	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.	Default
Unsigned Value	Value representing energy efficiency of conversion from DC to AC		M	full	%	0,4%	none
Communication:							
DP Address: (in the server)		Object Type: 1252		Property ID: 111			
		Start-Index: 1		N° of elements: 1			
Property access:		Read only ☒ -		Read/Write ☐			
Protection		Read level -		Write level -			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
When there is no conversion, the value shall be 0 (zero).							

4.5 New Data Point Types

4.5.1 DPT_Value_ReactivePower

<u>Format:</u>	4 octet: F ₃₂																				
octet nr.	1																				
field names	<table border="1"><tr><td>field 1 - 4</td></tr></table>					field 1 - 4															
field 1 - 4																					
encoding	<table border="1"><tr><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td></tr></table>					F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F						
<u>Encoding:</u>	The values are encoded in the IEE floating point format according to IEEE 754 single precision format.																				
<u>Unit:</u>	VAr																				
<u>Resol.:</u>	The resolution is given by the IEEE 754 format and varies by the used exponent.																				
<u>PDT:</u>	PDT_FLOAT																				
Datapoint Types																					
<u>ID:</u>	<u>Name:</u>	<u>Encoding:</u>	<u>Range:</u>	<u>Unit</u>	<u>Use:</u>																
14.081	DPT_Value_Reactive Power	see IEEE 754	see IEEE 754	VAR	G																

5 FB PCC Monitor

5.1 Aims and objectives

This Functional Block is designed to monitor the physical values at the Point of Common Coupling (PCC)

- Measure bidirectionally the electrical current of L1, L2, L3 at the PCC
- Measure the voltage of L1, L2, L3 at the PCC
- Calculate the present Power feed in the grid
- Calculate the present Power consumed from the grid
- Measure the frequency at the PCC

The PCC monitor measures the voltage, the current and optionally the frequency, at the grid connection. These values will be used by the EMS to support its optimization objectives, e.g. self-consumption. Optionally the values can be used to check if the CEM energy managed architecture behaves as planned.

5.2 Functional Block Diagram: PCC Monitor

Inputs	Point of Common Coupling Monitor		Output
	Parameter	Description	
		PowerFeedIn	PowerFeedIn
		PowerConsumption	PowerConsumption
		PowerEff	PowerEff
		VoltageL1	VoltageL1
		VoltageL2	VoltageL2
		VoltageL3	VoltageL3
		MainsFrequency	MainsFrequency
		CurrentL1	CurrentL1
		CurrentL2	CurrentL2
		CurrentL3	CurrentL3

Figure 6 – PCC Monitor

5.3 List of Datapoints: PCC Monitor

Outputs		
Datapoint	Description/Remarks	Datapoint Type
PowerFeedIn	Present power fed from the premises into the grid of the Distribution System Operator (DSO) (only positive values)	14.056 DPT_Value_Power
PowerConsumption	Present power sourced from the grid of the DSO (only positive values)	14.056 DPT_Value_Power
PowerEff	Present power, sourced or fed (fed is a negative value and sourced is a positive value) into the grid of the DSO	14.056 DPT_Value_Power
VoltageL1	AC Voltage of the output between phase L1 and N at the PCC	14.027 DPT_Value_Electric_Potential
VoltageL2	AC Voltage of the output between phase L2 and N at the PCC	14.027 DPT_Value_Electric_Potential
VoltageL3	AC Voltage of the output between phase L3 and N at the PCC	14.027 DPT_Value_Electric_Potential
MainsFrequency	Present Frequency value (Hz)	14.033 DPT_Value_Frequency
CurrentL1	AC Current of the output on phase L1 at the PCC	14.019 DPT_Value_Electric_Current
CurrentL2	AC Current of the output on phase L2 at the PCC	14.019 DPT_Value_Electric_Current
CurrentL3	AC Current of the output on phase L3 at the PCC	14.019 DPT_Value_Electric_Current

5.4 Detailed specification of the Datapoints

5.4.1 Inputs: PCC Monitor

This Functional Block has no inputs.

5.4.2 Outputs: PCC Monitor

5.4.2.1 PowerFeedIn

DP Name:	PowerFeedIn	Abbr.:		Mandatory	☒
FB Name:	PCC Monitor			Can be internal	☐
Description					
Present power fed from the premises into the grid of the Distribution System Operator (DSO) (only positive values)					
Datapoint Type					
DPT Name:					
DPT Format:	DPT Value Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float Value	Value representing the present power fed into the grid	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	51	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

5.4.2.2 PowerConsumption

DP Name:	PowerConsumption	Abbr.:		Mandatory	☒
FB Name:	PCC Monitor			Can be internal	☐
Description					
Present power which is fed from the premises into the grid of Distribution System Operator (DSO) (only positive values)					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float Value	Value representing the present power fed into the grid	M	full	W	none
Access Type					
◆ Output					
	this → M	☒	this → 1	☐	
	Spontaneous	☒	COV:	☒	Δ-Value: >1 %
			Cyclic	☐	Period:
	Request	☒			
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1253	Property ID:	52
		Start-Index:	1	N° of elements:	1
	Property access:	Read only	☒	Read/Write	☐
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

5.4.2.3 PowerEff

DP Name:	PowerEff	Abbr.:		Mandatory	☒																				
FB Name:	PCC Monitor			Can be internal	☐																				
Description																									
Present power, sourced or fed (fed is a negative value and sourced is a positive value) into the grid of DSO																									
Datapoint Type																									
DPT Name:																									
DPT Format:	DPT_Value_Power	DPT_ID:	14.056																						
Field		Supp.	Range	Unit	Default																				
Float Value	Value representing the present power sourced or fed from the grid	M	full	W	none																				
Access Type																									
◆ Output																									
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td>Request</td> <td>☒</td> <td>Period:</td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐	Request	☒	Period:						
this → M	☒	this → 1	☐																						
Spontaneous	☒	COV:	☒																						
		Δ-Value:	>1 %																						
		Cyclic	☐																						
Request	☒	Period:																							
Communication Type																									
◆ Group Object Datapoint					Mandatory: ☒																				
Default Group Address:		none																							
◆ Property					Mandatory: ☒																				
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	53																					
	Start-Index:	1	N° of elements:	1																					
Property access:	Read only	☒	Read/Write	☐																					
Protection *)	Read level	-	Write level	-																					
Dynamics																									
Power down:	Save:	☐																							
Power up:	Value:	No initialisation:	☒	Default value:	☐																				
		Saved value:	☐	Current value (not for input):	☐																				
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																				
Exception Handling																									
Special Features																									

5.4.2.4 VoltageL1

DP Name:	VoltageL1		Abbr.:		Mandatory	☒
FB Name:	PCC Monitor				Can be internal	☐
Description						
AC Voltage of the output between phase L1 and N at the PCC						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Electric_Potential			DPT_ID:	14.027	
Field				Supp.	Range	Unit
Float Value	AC voltage between L1 and N			M	full	V
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	54		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

5.4.2.5 VoltageL2

DP Name:	VoltageL2		Abbr.:		Mandatory	☒																								
FB Name:	PCC Monitor				Can be internal	☐																								
Description																														
AC Voltage of the output between phase L2 and N at the PCC																														
Datapoint Type																														
DPT Name:																														
DPT Format:	DPT_Value_Electric_Potential			DPT_ID:	14.027																									
Field		Supp.	Range	Unit	Default																									
Float Value	AC voltage between L2 and N			M	full	V none																								
Access Type																														
◆ Output																														
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒								
this → M	☒	this → 1	☐																											
Spontaneous	☒	COV:	☒																											
		Δ-Value:	>1 %																											
		Cyclic	☐																											
		Period:																												
Request	☒																													
Communication Type																														
◆ Group Object Datapoint					Mandatory:	☒																								
Default Group Address:		none																												
◆ Property					Mandatory:	☒																								
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	55																										
	Start-Index:	1	N° of elements:	1																										
Property access:	Read only	☒	Read/Write	☐																										
Protection *)	Read level	-	Write level	-																										
Dynamics																														
Power down:	Save:	☐																												
Power up:	Value:	No initialisation:	☒	Default value:	☐																									
		Saved value:	☐	Current value (not for input):	☐																									
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																									
Exception Handling																														
Special Features																														

5.4.2.6 VoltageL3

DP Name:	VoltageL3		Abbr.:		Mandatory	☒
FB Name:	PCC Monitor				Can be internal	☐
Description						
AC Voltage of the output between phase L3 and N at the PCC						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Electric_Potential			DPT_ID:	14.027	
Field				Supp.	Range	Unit
Float Value	AC voltage between L3 and N			M	full	V
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	56		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

5.4.2.7 MainsFrequency

DP Name:	MainsFrequency	Abbr.:		Mandatory	☐
FB Name:	PCC Monitor			Can be internal	☐
Description					
Present Frequency value (Hz)					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Frequency	DPT_ID:	14.033		
Field		Supp.	Range	Unit	Default
Float Value	Value representing present frequency	M	full	Hz	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	57	
	Start-Index:	1	N° of elements:	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

5.4.2.8 CurrentL1

DP Name:	CurrentL1		Abbr.:		Mandatory	☒
FB Name:	PCC Monitor				Can be internal	☐
Description						
AC Current of the output on phase L1 at the PCC						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Electric_Current			DPT_ID:	14.019	
Field				Supp.	Range	Unit
Float Value	Value representing the AC current on phase L1			M	full	A
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	58		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

5.4.2.9 CurrentL2

DP Name:	CurrentL2		Abbr.:		Mandatory	☒
FB Name:	PCC Monitor				Can be internal	☐
Description						
AC Current of the output on phase L2 at the PCC						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Electric_Current			DPT_ID:	14.019	
Field				Supp.	Range	Unit
Float Value	Value representing the AC current on phase L1			M	full	A
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	59		
	Start-Index:	1	N° of elements:	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

5.4.2.10 CurrentL3

DP Name:	CurrentL3	Abbr.:		Mandatory	☒																									
FB Name:	PCC Monitor			Can be internal	☐																									
Description																														
AC Current of the output on phase L3 at the PCC																														
Datapoint Type																														
DPT Name:																														
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019																											
Field		Supp.	Range	Unit	Default																									
Float Value	Value representing the AC current on phase L1	M	full	A	none																									
Access Type																														
◆ Output																														
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td colspan="2"></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒								
this → M	☒	this → 1	☐																											
Spontaneous	☒	COV:	☒																											
		Δ-Value:	>1 %																											
		Cyclic	☐																											
		Period:																												
Request	☒																													
Communication Type																														
◆ Group Object Datapoint				Mandatory:	☒																									
Default Group Address:		none																												
◆ Property				Mandatory:	☒																									
DP Address: (in the server)	IO Type(ID):	1253	Property ID:	60																										
	Start-Index:	1	N° of elements:	1																										
Property access:	Read only	☒	Read/Write	☐																										
Protection *)	Read level	-	Write level	-																										
Dynamics																														
Power down:	Save:	☐																												
Power up:	Value:	No initialisation:	☒	Default value:	☐																									
		Saved value:	☐	Current value (not for input):	☐																									
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																									
Exception Handling																														
Special Features																														

5.4.3 Parameter: PCC Monitor

This Functional Block has no parameters.

5.4.4 Diagnostic Data: PCC Monitor

This Functional Block has no diagnostic data.